Yojun Moon (문요준)

yojun313@postech.ac.kr github.com/yojun313 linkedin.com/in/yojun-moon www.yojun.dev

Education

POSTECH (Pohang University of Science and Technology) *Mar 2023 – Present*

B.S. in Computer Science and Engineering

Work Experience

Samsung Fire & Marine Insurance *Jul 2026 – Aug 2026*

AI Engineer Intern, AI Development Part

Developed two computer vision modules for insurance claim data, from model training and evaluation to web-based processing services.

- License plate masking — designed a vehicle-to-plate detection pipeline, achieving 99.0% recall on the image evaluation set and 27+ fps in the development GPU environment.
- Document forgery detection — built FinForge with 6,000 original/forged images using eight generative AI models, fine-tuned TruFor, and raised document-level test accuracy from about 55% to 93%.
- Evaluated blur, noise, JPEG recompression, and external data; organized reports, code, and datasets for follow-up research.
- Skills: Python, PyTorch, Ultralytics YOLO, OpenCV, FFmpeg, FastAPI

Korean National Police University – FPEI (Future Public Safety Engineering Institute) *Nov 2023 – Present*

Research System Developer

Developed and operate the institute's research platform, extending desktop workflows to production web services at knpu.re.kr/systems (github.com/yojun313/knpu).

- Data collection — built CRAWLER with asynchronous queues and proxy IP rotation for Naver News, Blog, Cafe, and YouTube, with shared job monitoring.
- Research tools — implemented STATISTICS, NETWORK, and KEMKIM analysis workflows with AI interpretation, unified by MANAGER; integrated GPU audio transcription through WHISPER.
- Agent chatbot — implemented a tool-selection and execution loop with BM25 retrieval, source inspection, public/member permissions, and execution limits.
- AI complaint drafting — combined conversational fact collection, statute and precedent retrieval, element analysis, drafting, and rule checks plus LLM review; generated editable Word/PDF documents.
- Server and LLM operations — solely manage two research servers, including a dual-A100 80GB GPU server; connected them through SSH tunneling and run gpt-oss-120b via vLLM as the shared base LLM, managed through vllm-manager.
- Received a Letter of Appreciation from the Korean National Police University for research software contributions (Apr 2024).
- Skills: Python, FastAPI, MongoDB, Qt, Linux, vLLM, Nginx, PM2, WebSocket, BM25

POSTECH HAiV Lab *Jan 2026 – Feb 2026*

Undergraduate Researcher

Contributed to ACCIDENTOR, a traffic-accident analysis framework combining local vision-language models with embedding-based retrieval from a vector database to support fault-ratio estimation from dashcam video.

- Skills: Python, PyTorch, Vision-Language Models, embeddings, vector database

Startup Team (founded by a POSTECH CSE alumnus) *Feb 2025 – Jul 2025*

Backend Developer Intern

Developed the backend of a social networking service and supported data-driven product decisions.

- Built the SNS backend with NestJS, developed web crawlers, and analyzed collected data.
- Completed the 2025 I-Corps program, including three weeks of entrepreneurship training at UC Berkeley (KIC Tech Frontier Program).
- Skills: NestJS, TypeScript, Python, MongoDB, Crawler

Projects

PCSS (POSTECH Computer Scientist Search) *Jan 2025 – Mar 2026*

Commissioned faculty candidate search service for the POSTECH CSE department.

- Parsed DBLP XML into MongoDB and used the Grok API for Korean-name scoring, reducing manual faculty-candidate browsing from hours to queries taking seconds.
- Built conference and name-score management, researcher profiles/rankings, monthly updates, and email subscriptions.
- Developing NameClassifier with LoRA adapters and a scalar scoring head on Qwen3-0.6B-Base, using approximately 4.7 million name records; achieved 97.69% F1 on a 30,550-record benchmark at a 0.5 threshold.
- Built a FastAPI inference/evaluation interface with SQLite history, threshold adjustment, and error inspection.
- Deployed at pcss.postech.ac.kr / name-classifier.yojun.dev
- Skills: Python, FastAPI, MongoDB, Grok API, PyTorch, Transformers, LoRA, DDP, SQLite

Commeet *Aug 2026 – Present*

Scheduling and consultation record service for POSTECH CSE professors, with 100+ cumulative users.

- Maximized conflict-free assignments with min-cost max-flow, preferred-slot costs, and session regrouping.
- Added guest responses, a reusable contact book, email-domain verification, and Google Calendar integration.
- Built encrypted professor consultation records, student notes, signatures, printable forms, and guidance-fee receipts.
- Deployed at commeet.postech.ac.kr
- Skills: Python, FastAPI, MongoDB (Motor), Jinja2, JavaScript, Google Calendar API, Fernet

POPO *Jan 2026 – Feb 2026*

Contributed to the backend of the campus web service operated by PoApper, POSTECH's student developer club.

- Skills: NestJS, TypeScript

Activities

POSTECH Broadcasting Station (PBS) *Mar 2023 – Dec 2025*

Member of the Programming & Production Department.

- Filmed, edited, and live-broadcast campus entertainment programs.

Programming Tutor *Jan 2025 – Present*

Teaching Python and C to middle and high school students.

- Topics include object-oriented programming, MongoDB, and FastAPI.

Skills

Programming Languages	Python, C, C++, TypeScript, JavaScript
AI / Machine Learning	PyTorch, Transformers, LoRA, DDP, YOLO, OpenCV, vLLM, LLM/VLM APIs
Backend	FastAPI, NestJS, MongoDB, SQLite, REST APIs, WebSocket
Tools / Miscellaneous	Docker, Git, Linux, Nginx, PM2, SSH, Qt, FFmpeg